How to Learn as Prospective Sophomores

Mainly in Math & English

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Outline

1. About me
2. Motivation
3. Learning skills
4. About CS
5. Contacts
6. Q&A

CV

Education:

1. B.S. 201609–202006 CS, CCSE, HNNU
2. Ph.D 202009–Now CS, SCSE, BUAA

Homepage:

1. Github: https://zhoukanglei.github.io/
2. Google scholar: https://scholar.google.com/citations?user=yG1FaioAAAAJ

Figure 1: BUAA

What is learning

Definition (From Google search)

Exploring is to travel through the unknown to learn or discover new information. There's much learning and great power in exploring the unknown.

Definition (My view)

Learning is exploring the unknown, and deep learning is exploring the law.

Figure 2: Most Fun Multiplayer Games — Zynga

Why this topic

1. IMPORTANT
2. In critical phase of learning for sophomores

Figure 3: In Blended Learning Blog $n°10$

Why do we need to learn [1]

1. Learning is essential to our existence.
2. To live a life without continuous learning is unthinkable.
3. The only thing that is constant is change.

[1] https://www.linkedin.com/pulse/7-reasons-why-continuous-learning-important-amit-nagpal

Why is English so significant

From Google

Knowing English increases your chances of getting a good job in a multinational company within your home country or for finding work abroad. It's also the language of international communication, the media and the Internet, so learning English is important for socializing and entertainment as well as work!

My view

Working or Researching, English is a key to open the door to the world for us.

Figure 4: Recommend Google (http://googlehelper.net/)

About English

101 Ways to Learn English [2]

1. Don't be afraid to make mistakes. ...
2. Surround yourself in English. ...
3. Practise every day. ...
4. Tell your family and friends about your study plan. ...
5. Practise the 4 core skills: reading, writing, speaking and listening.

[2] https://www.ecenglish.com/en/101-ways-to-learn-english

Why is Math so significant

From Google

Math helps us have better problem-solving skills. Math helps us think analytically and have better reasoning abilities. Analytical thinking refers to the ability to think critically about the world around us. ... Analytical and reasoning skills are essential because they help us solve problems and look for solutions.

My view

For us students major in CS, Math is the foundation.

Figure 5: Algorithm & Math thinking

About math

How to Study Maths: 7 Tips for Problem Solving [3]:

1. Practice, Practice & More Practice. It is impossible to study maths properly by just reading and listening. ...
2. Review Errors. ...
3. Master the Key Concepts. ...
4. Understand your Doubts. ...
5. Create a Distraction Free Study Environment. ...
6. Create a Mathematical Dictionary. ...
7. Apply Maths to Real World Problems.

[3] https://www.goconqr.com/en/examtime/blog/how-to-study-maths/

Be professional

Code

```
# Define parameters for the walk
step_n = 10000

# Simulate steps in 1D
step_shape = (step_n,dims)
```

Math symbols:

1. Variables: a, b
2. Vectors: $\boldsymbol{w}, \boldsymbol{\phi}$
3. Matrices: $\boldsymbol{\Phi}, \boldsymbol{\Psi}$
4. Others: $\mathbb{R}, \arg\max$

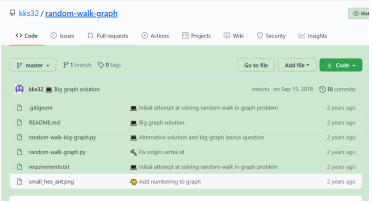

Figure 6: A repository in Github

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Figure 7: QQ groups

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Figure 8: WeChat subscription

Thank you!